SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	Ch.saisri	**Reg. No.:**	192472367
Section / Batch:	Cse-ai	**Date:**	

Code of Conduct
I ch,saisri (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: ___________________________

Instructions
- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression Possible Antecedents Valid/Invalid Reason

d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. **Maintain entity coherence using assignment, mistakes, and you.**
2. **Use a Cause–Effect or Elaboration discourse relation.**
3. **Include at least two keywords from:**
    - **review**
    - **improve**
    - **confident**
    - **feedback**

4. **Response length must be 2–3 sentences.**
5. **Maintain a positive and polite tone.**
6. **Do not give contradictory advice.**

Tasks:

a) Identify the dialog acts required.

b) Identify the important entities that must be maintained.

c) Generate three possible responses satisfying all constraints.

d) Compare the responses based on:

- **Coherence**
- **Constraint satisfaction**
- **Politeness**
- **Relevance**

e) Select the best response and justify your answer.

f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- **A financial institution**
- **The side of a river**

Constraints:

1. **Resolve the correct word sense using contextual information.**
2. **Identify important entities and actions.**
3. **Represent the meaning using predicate logic.**
4. **Preserve semantic relationships.**
5. **Maintain coherence in the generated paraphrase.**

Tasks:

a) Identify the possible meanings of bank.

b) Select the correct sense and justify your decision using contextual constraints.

c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

QUESTION 1 — Reference Resolution and Discourse Coherence

```
# QUESTION 1: REFERENCE RESOLUTION

print("=" * 65)
print("REFERENCE RESOLUTION AND DISCOURSE COHERENCE")
print("=" * 65)

paragraph = """
Ravi met Arun at the library.
He was looking for a book on Artificial Intelligence.
Arun helped him find it.
Later, they discussed the book before leaving.
"""

print("\nParagraph:")
print(paragraph)


# -------------------------------------------------
# Referring Expressions
# -------------------------------------------------

references = {
    "He": ["Ravi", "Arun"],
    "him": ["Ravi", "Arun"],
    "it": ["book"],
    "they": ["Ravi", "Arun"],
    "the book": ["book"]
}

print("\n" + "=" * 65)
print("REFERRING EXPRESSIONS")
print("=" * 65)
```

```
for expression, antecedents in references.items():

    print("\nReferring Expression:", expression)
    print("Possible Antecedents:", antecedents)


# -----------------------------------------------
# Constraint Analysis
# -----------------------------------------------

print("\n" + "=" * 65)
print("CONSTRAINT ANALYSIS")
print("=" * 65)

print("""
1. 'He'
   Possible: Ravi, Arun
   Final: Ravi
   Reason: Recency and grammatical context.

2. 'him'
   Possible: Ravi, Arun
   Final: Ravi
   Reason: Arun helped Ravi find the book.

3. 'it'
   Possible: book
   Final: book
   Reason: 'it' refers to the previously mentioned book.

4. 'they'
   Possible: Ravi, Arun
   Final: Ravi and Arun
   Reason: Plural pronoun refers to both people.

5. 'the book'
   Final: Previously mentioned book.
   Reason: Semantic and discourse coherence.
""")


# -----------------------------------------------
# Coreference Chains
# -----------------------------------------------

print("=" * 65)
print("FINAL COREFERENCE CHAINS")
print("=" * 65)

print("Ravi -> He -> him -> they")
print("Arun -> they")
print("book -> it -> the book")
```

```
# -------------------------------------------------
# Rewritten Paragraph
# -------------------------------------------------

print("\n" + "=" * 65)
print("REWRITTEN PARAGRAPH")
print("=" * 65)

rewritten = """
Ravi met Arun at the library.
Ravi was looking for a book on Artificial Intelligence.
Arun helped Ravi find the book.
Later, Ravi and Arun discussed the book before leaving.
"""

print(rewritten)


# -------------------------------------------------
# Ambiguity
# -------------------------------------------------

print("=" * 65)
print("AMBIGUITY ANALYSIS")
print("=" * 65)

print("""
If 'He was looking for a book' appears before Arun
is introduced, the pronoun 'He' has no clear antecedent.

Example:
Ravi met someone at the library.
He was looking for a book.
Arun arrived later.

The system may not know who 'He' refers to.
This reduces discourse coherence.
""")

print("RESULT:")
print("Reference resolution uses gender, number, recency,")
print("grammatical role, semantic compatibility and context.")
```

output:

```
=================================================================
REFERENCE RESOLUTION AND DISCOURSE COHERENCE
=================================================================

Paragraph:

Ravi met Arun at the library.
He was looking for a book on Artificial Intelligence.
Arun helped him find it.
Later, they discussed the book before leaving.


=================================================================
REFERRING EXPRESSIONS
=================================================================

Referring Expression: He
Possible Antecedents: ['Ravi', 'Arun']

Referring Expression: him
Possible Antecedents: ['Ravi', 'Arun']

Referring Expression: it
Possible Antecedents: ['book']

Referring Expression: they
Possible Antecedents: ['Ravi', 'Arun']

Referring Expression: the book
Possible Antecedents: ['book']

=================================================================
CONSTRAINT ANALYSIS
=================================================================

1. 'He'
   Possible: Ravi, Arun
   Final: Ravi
   Reason: Recency and grammatical context.

2. 'him'
```

```
2. 'him'
   Possible: Ravi, Arun
   Final: Ravi
   Reason: Arun helped Ravi find the book.

3. 'it'
   Possible: book
   Final: book
   Reason: 'it' refers to the previously mentioned book.

4. 'they'
   Possible: Ravi, Arun
   Final: Ravi and Arun
   Reason: Plural pronoun refers to both people.

5. 'the book'
   Final: Previously mentioned book.
   Reason: Semantic and discourse coherence.

=================================================================
FINAL COREFERENCE CHAINS
=================================================================
Ravi -> He -> him -> they
Arun -> they
book -> it -> the book

=================================================================
REWRITTEN PARAGRAPH
=================================================================

Ravi met Arun at the library.
Ravi was looking for a book on Artificial Intelligence.
Arun helped Ravi find the book.
Later, Ravi and Arun discussed the book before leaving.
=================================================================
AMBIGUITY ANALYSIS
=================================================================

If 'He was looking for a book' appears before Arun
is introduced, the pronoun 'He' has no clear antecedent.

Example:
Ravi met someone at the library.
He was looking for a book.
Arun arrived later.

The system may not know who 'He' refers to.
This reduces discourse coherence.

RESULT:
Reference resolution uses gender, number, recency,
grammatical role, semantic compatibility and context.
```

Important final chains
Ravi → He → him
Arun → they
Ravi + Arun → they
book → it → the book

QUESTION 2 — Dialog Act Recognition and Response Generation

This code generates **3 responses**, checks the required constraints, and selects the best one.

```
# QUESTION 2: DIALOG ACT RECOGNITION

print("=" * 65)
print("DIALOG ACT RECOGNITION AND RESPONSE GENERATION")
print("=" * 65)


student = (
    "I submitted my assignment, but I am worried "
    "that I may have made many mistakes."
)

print("\nStudent:")
print(student)


# --------------------------------------------------
# Dialog Acts
# --------------------------------------------------

print("\n" + "=" * 65)
print("DIALOG ACTS")
print("=" * 65)

print("1. REASSURE")
print("2. ADVISE")


# --------------------------------------------------
# Entities
# --------------------------------------------------

print("\n" + "=" * 65)
print("IMPORTANT ENTITIES")
print("=" * 65)

print("assignment")
print("mistakes")
print("you")


# --------------------------------------------------
# Three Responses
```

```
# ------------------------------------------------

responses = [

    "You can feel confident about your assignment, and mistakes are a normal part of learning. "
    "Review your work carefully and use the feedback to improve your future assignments.",

    "Don't worry, you have already completed your assignment and you can learn from any mistakes. "
    "Review the work and use feedback to improve your understanding.",

    "You should not worry too much because your assignment is a learning opportunity. "
    "Review your answers and use feedback to become more confident and improve."
]


print("\n" + "=" * 65)
print("THREE POSSIBLE RESPONSES")
print("=" * 65)

for i, response in enumerate(responses, 1):

    print("\nResponse", i + 0)
    print(response)


# ------------------------------------------------
# Constraint Checking
# ------------------------------------------------

print("\n" + "=" * 65)
print("CONSTRAINT CHECKING")
print("=" * 65)

for i, response in enumerate(responses, 1):

    keywords = [
        "review",
        "improve",
        "confident",
        "feedback"
    ]

    count = 0

    for word in keywords:

        if word in response.lower():
            count += 1

    sentences = response.count(".")

    print("\nResponse", i)
    print("Required keywords found:", count)
```

```python
    print("Sentences:", sentences)

    if count >= 2 and 2 <= sentences <= 3:
        print("Constraint Status: SATISFIED")
    else:
        print("Constraint Status: NOT SATISFIED")


# ---------------------------------------------------
# Comparison
# ---------------------------------------------------

print("\n" + "=" * 65)
print("COMPARISON")
print("=" * 65)

print("""
Response 1:
Coherence     : High
Politeness    : High
Relevance     : High
Constraints   : Satisfied

Response 2:
Coherence     : High
Politeness    : High
Relevance     : High
Constraints   : Satisfied

Response 3:
Coherence     : High
Politeness    : High
Relevance     : High
Constraints   : Satisfied
""")


# ---------------------------------------------------
# Best Response
# ---------------------------------------------------

print("=" * 65)
print("BEST RESPONSE")
print("=" * 65)

print(responses[0])

print("\nReason:")
print("It directly reassures the student and provides useful advice.")
print("It maintains the assignment and mistakes entities.")
print("It uses review, feedback, improve and confident.")
print("It is positive, polite and coherent.")
```

```python
# ---------------------------------------------------
# Entity Coherence
# ---------------------------------------------------

print("\n" + "=" * 65)
print("ENTITY COHERENCE")
print("=" * 65)

print("""
If entity coherence is not maintained, the chatbot may
confuse the assignment with another task or give advice
unrelated to the student's mistakes.

This makes the response unclear and reduces relevance.
""")

print("RESULT:")
print("The best response performs both REASSURE and ADVISE.")
```

output:

```
======================================================================
DIALOG ACT RECOGNITION AND RESPONSE GENERATION
======================================================================

Student:
I submitted my assignment, but I am worried that I may have made many mistakes.

======================================================================
DIALOG ACTS
======================================================================
1. REASSURE
2. ADVISE

======================================================================
IMPORTANT ENTITIES
======================================================================
assignment
mistakes
you

======================================================================
THREE POSSIBLE RESPONSES
======================================================================

Response 1
You can feel confident about your assignment, and mistakes are a normal part of learning. Review your work carefully and use the feedback to improve your
future assignments.

Response 2
Don't worry, you have already completed your assignment and you can learn from any mistakes. Review the work and use feedback to improve your understandi
ng.

Response 3
You should not worry too much because your assignment is a learning opportunity. Review your answers and use feedback to become more confident and improv
e.

======================================================================
CONSTRAINT CHECKING
======================================================================

Response 1
```

```
======================================================================
CONSTRAINT CHECKING
======================================================================

Response 1
Required keywords found: 4
Sentences: 2
Constraint Status: SATISFIED

Response 2
Required keywords found: 3
Sentences: 2
Constraint Status: SATISFIED

Response 3
Required keywords found: 4
Sentences: 2
Constraint Status: SATISFIED

======================================================================
COMPARISON
======================================================================

Response 1:
Coherence      : High
Politeness     : High
Relevance      : High
Constraints    : Satisfied

Response 2:
Coherence      : High
Politeness     : High
Relevance      : High
Constraints    : Satisfied

Response 3:
Coherence      : High
Politeness     : High
Relevance      : High
Constraints    : Satisfied
```

```
======================================================================
BEST RESPONSE
======================================================================
You can feel confident about your assignment, and mistakes are a normal part of learning. Review your work carefully and use the feedback to improve your
future assignments.

Reason:
It directly reassures the student and provides useful advice.
It maintains the assignment and mistakes entities.
It uses review, feedback, improve and confident.
It is positive, polite and coherent.

======================================================================
ENTITY COHERENCE
======================================================================

If entity coherence is not maintained, the chatbot may
confuse the assignment with another task or give advice
unrelated to the student's mistakes.

This makes the response unclear and reduces relevance.

RESULT:
The best response performs both REASSURE and ADVISE.
```

Best response

You can feel confident about your assignment, and mistakes are a normal part of learning. Review your work carefully and use the feedback to improve your future assignments.

It satisfies the **2-sentence requirement**, maintains the entities, uses multiple required keywords, and has a positive tone.

QUESTION 3 — Word Sense Disambiguation and Meaning Representation

```
# QUESTION 3: WORD SENSE DISAMBIGUATION

print("=" * 65)
print("WORD SENSE DISAMBIGUATION")
print("=" * 65)


# ------------------------------------------------
# Source Sentence
# ------------------------------------------------

sentence = (
    "Priya sat on the bank and watched the boats "
    "moving across the water."
)

print("\nSource Sentence:")
print(sentence)


# ------------------------------------------------
# Possible meanings
# ------------------------------------------------

print("\n" + "=" * 65)
print("POSSIBLE MEANINGS OF 'BANK'")
print("=" * 65)

print("Sense 1: Bank = Financial Institution")
print("Sense 2: Bank = Side of a River")


# ------------------------------------------------
# Context Analysis
# ------------------------------------------------

print("\n" + "=" * 65)
print("CONTEXTUAL ANALYSIS")
print("=" * 65)

context_words = [
    "sat",
    "boats",
    "moving",
    "water"
]

print("Important context words:")
```

```
print(context_words)

print("\nThese words are strongly related to a river.")
print("Therefore, 'bank' means the side of a river.")


# --------------------------------------------------
# Word Sense Decision
# --------------------------------------------------

print("\n" + "=" * 65)
print("FINAL WORD SENSE")
print("=" * 65)

print("bank -> SIDE OF A RIVER")


# --------------------------------------------------
# Entities and Actions
# --------------------------------------------------

print("\n" + "=" * 65)
print("ENTITIES AND ACTIONS")
print("=" * 65)

print("Entity 1: Priya")
print("Entity 2: River bank")
print("Entity 3: Boats")
print("Entity 4: Water")

print("\nAction 1: sat")
print("Action 2: watched")
print("Action 3: moving")


# --------------------------------------------------
# Predicate Logic
# --------------------------------------------------

print("\n" + "=" * 65)
print("PREDICATE LOGIC REPRESENTATION")
print("=" * 65)

print("""
sit(Priya)
bank(RiverBank)
beside(Priya, RiverBank)

watch(Priya, Boats)
boat(Boats)

move(Boats)
water(Water)
```

```
across(Boats, Water)
""")


# --------------------------------------------------
# Simple logical representation
# --------------------------------------------------

print("Combined Representation:")

logic = (
    "sit(Priya) AND "
    "bank(RiverBank) AND "
    "beside(Priya,RiverBank) AND "
    "watch(Priya,Boats) AND "
    "boat(Boats) AND "
    "move(Boats) AND "
    "water(Water) AND "
    "across(Boats,Water)"
)

print(logic)


# --------------------------------------------------
# Unambiguous Paraphrase
# --------------------------------------------------

print("\n" + "=" * 65)
print("UNAMBIGUOUS PARAPHRASE")
print("=" * 65)

paraphrase = (
    "Priya sat beside the river and watched the boats "
    "moving across the water."
)

print(paraphrase)


# --------------------------------------------------
# WSD and Machine Translation
# --------------------------------------------------

print("\n" + "=" * 65)
print("WSD AND MACHINE TRANSLATION")
print("=" * 65)

print("""
Word Sense Disambiguation identifies the correct meaning
of an ambiguous word before translation.
```

```
For example:

bank = financial institution
OR
bank = river side

Correct sense selection prevents incorrect translation
and produces a more meaningful target sentence.
""")


print("RESULT:")
print("The word 'bank' means the side of a river.")
print("Context such as boats, water and moving helps")
print("the system select the correct sense.")
```

output:

```
=================================================================
DIALOG ACT RECOGNITION AND RESPONSE GENERATION
=================================================================

Student:
I submitted my assignment, but I am worried that I may have made many mistakes.

=================================================================
DIALOG ACTS
=================================================================
1. REASSURE
2. ADVISE

=================================================================
IMPORTANT ENTITIES
=================================================================
assignment
mistakes
you

=================================================================
THREE POSSIBLE RESPONSES
=================================================================

Response 1
You can feel confident about your assignment, and mistakes are a normal part of learning. Review your work carefully and use the feedback to improve your
future assignments.

Response 2
Don't worry, you have already completed your assignment and you can learn from any mistakes. Review the work and use feedback to improve your understandi
ng.

Response 3
You should not worry too much because your assignment is a learning opportunity. Review your answers and use feedback to become more confident and improv
e.

=================================================================
CONSTRAINT CHECKING
=================================================================
```

```
=================================================================
CONSTRAINT CHECKING
=================================================================

Response 1
Required keywords found: 4
Sentences: 2
Constraint Status: SATISFIED

Response 2
Required keywords found: 3
Sentences: 2
Constraint Status: SATISFIED

Response 3
Required keywords found: 4
Sentences: 2
Constraint Status: SATISFIED

=================================================================
COMPARISON
=================================================================

Response 1:
Coherence     : High
Politeness    : High
Relevance     : High
Constraints   : Satisfied

Response 2:
Coherence     : High
Politeness    : High
Relevance     : High
Constraints   : Satisfied

Response 3:
Coherence     : High
Politeness    : High
Relevance     : High
```

```
Relevance     : High
Constraints   : Satisfied

======================================================================
BEST RESPONSE
======================================================================
You can feel confident about your assignment, and mistakes are a normal part of learning. Review your work carefully and use the feedback to improve your
future assignments.

Reason:
It directly reassures the student and provides useful advice.
It maintains the assignment and mistakes entities.
It uses review, feedback, improve and confident.
It is positive, polite and coherent.

======================================================================
ENTITY COHERENCE
======================================================================

If entity coherence is not maintained, the chatbot may
confuse the assignment with another task or give advice
unrelated to the student's mistakes.

This makes the response unclear and reduces relevance.

RESULT:
The best response performs both REASSURE and ADVISE.
```

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	3	4	4	4	3	30	90
2	3	3	4	4	4	30	
3	4	3	4	3	4	30	

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ______________________	Date: ______________________	Date: ______________________